

Problem Statement ID – SIH26076

Problem Statement Title – Personalised homepage for the Mausam mobile application

Theme – Smart Automation

PS Category – Software

Team ID – 195697

Team Name – Zencoderss

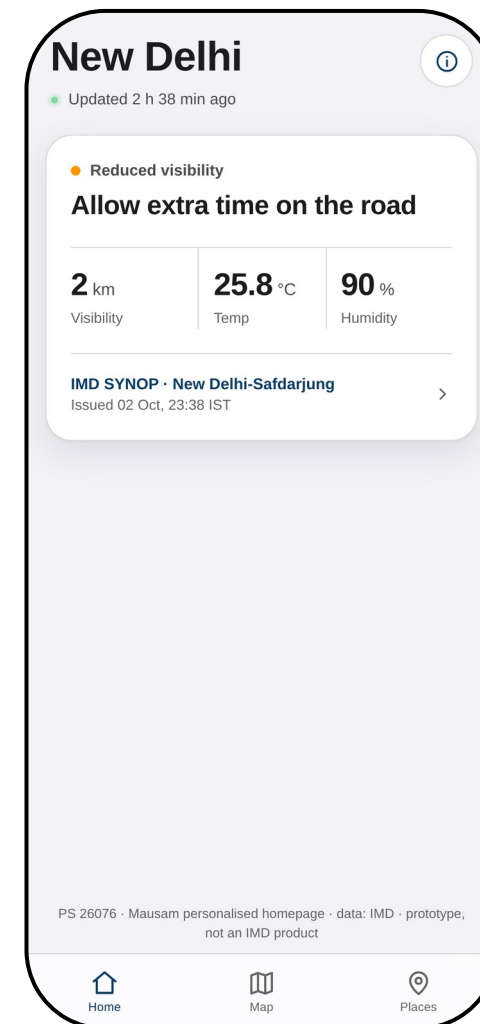
Organisation – Ministry of Earth Sciences (IMD)

Mausam home: actions with proof

Every number shows its source, station, issue time and age. Every advisory can be re-derived by hand.



LIVE DEMO <https://mausam-sih26076-devanshdhangar70-7520s-projects.vercel.app/>
Guest-first PWA · works offline · English and Hindi



WORKING PROTOTYPE · HOME (ILLUSTRATIVE)

Mausam home: actions with proof**Proposed Solution (Describe your Idea/Solution/Prototype)**

A home screen that adapts to you: map on top, action cards below.

Detailed explanation of the proposed solution

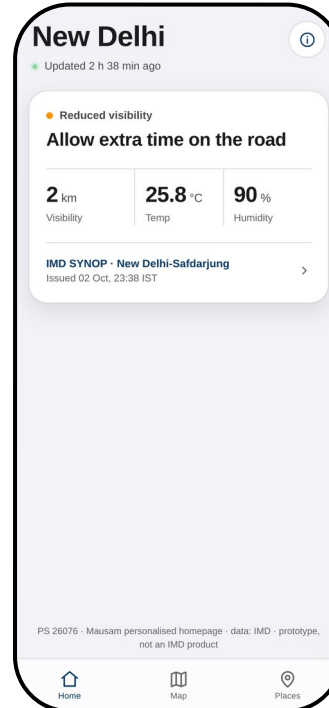
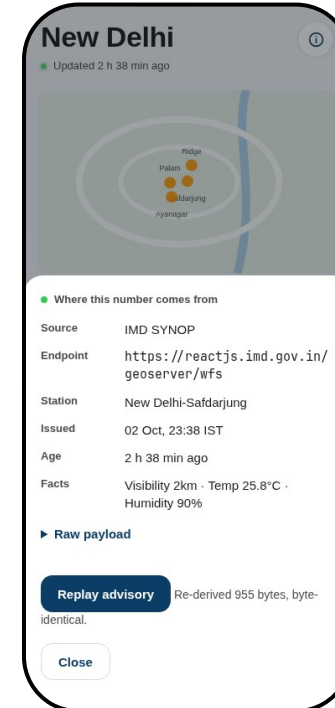
Onboarding picks your persona. The home reorders: the farmer sees sowing first, the commuter sees fog first.

How it addresses the problem

Every card says what to do, shows the readings behind it, and names its source, station and age.

Innovation and uniqueness of the solution

Source on every number. Replayable advice. Honest gaps where data is missing. A grounded bot that never invents numbers.

**HOME · MAP + ACTIONS****TAP ⓘ → PROVENANCE****Source on every number**

station + issue time + age on each reading; tap ⓘ for the full provenance sheet

Advice you can replay

same data → same card, byte-identical

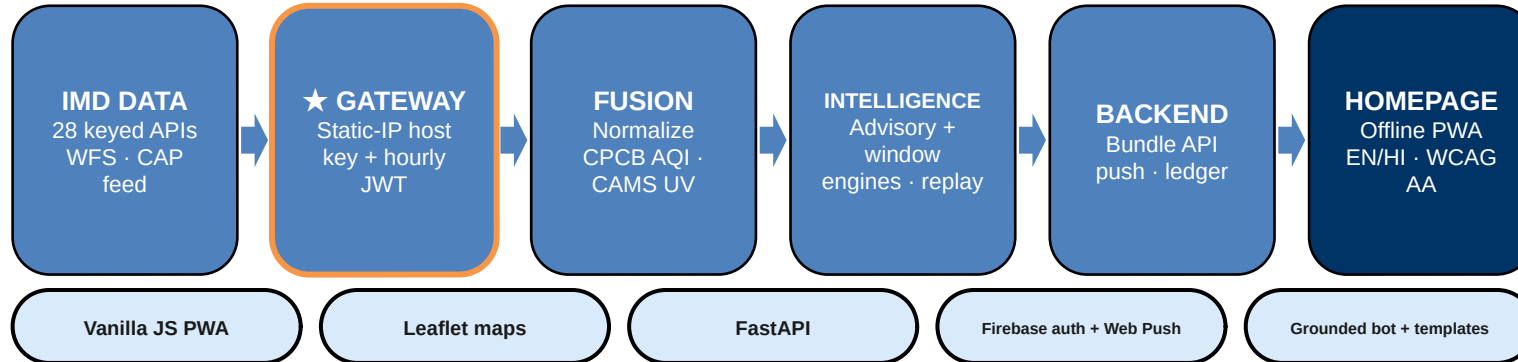
Honest gaps, no guesses

no data → a gap card, never a zero

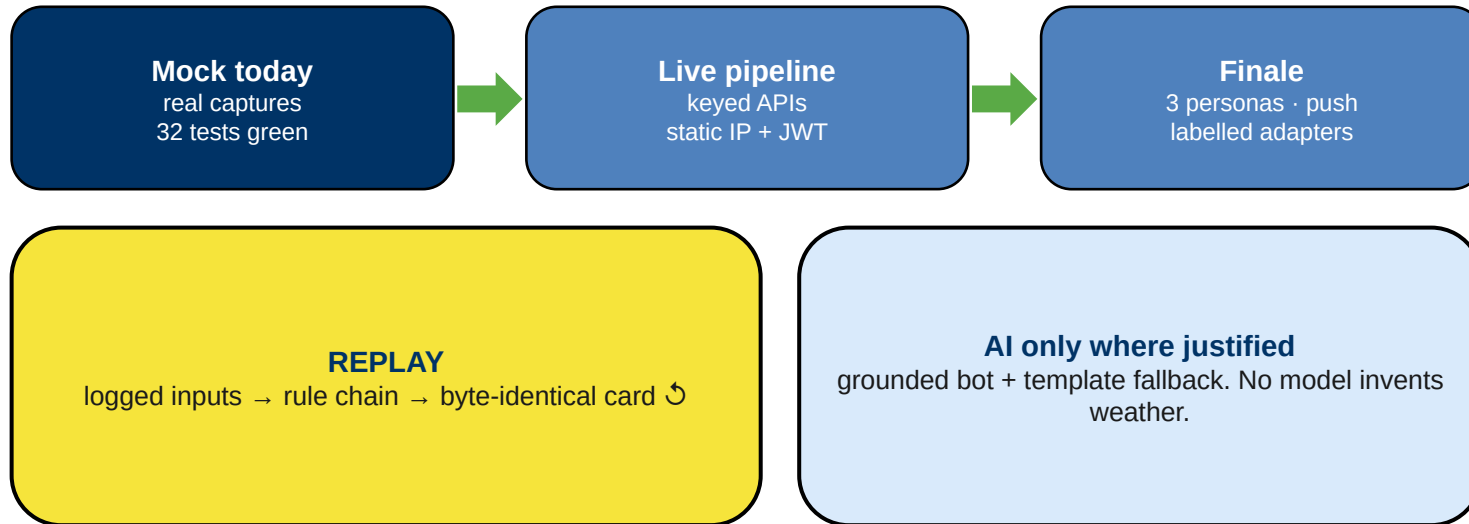
PERSONAS · Commuter · Farmer · Health

Every number on this screen can tell you where it came from, when it was issued, and how old it is. Every advisory can be re-derived by hand.

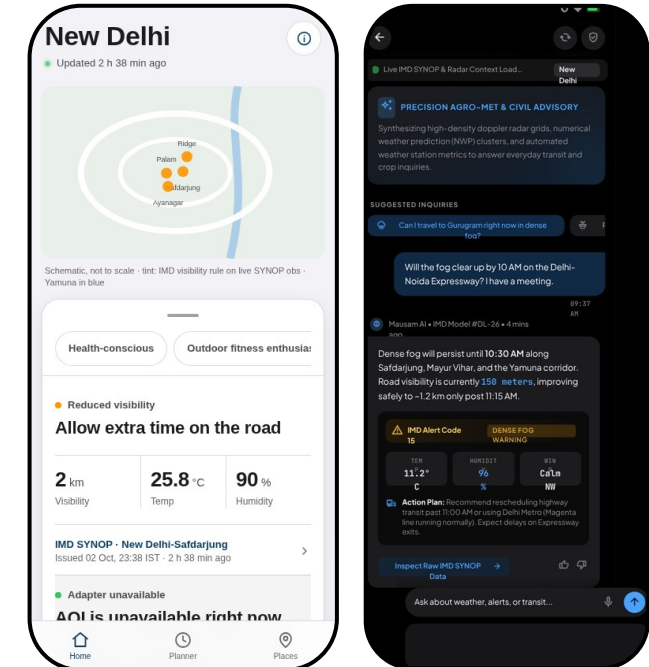
Technologies to be used (e.g. programming languages, frameworks, hardware)



Methodology and process for implementation (Flow Charts/Images/ working prototype)



WORKING PROTOTYPE · ASSISTANT DESIGN PREVIEW



BYTE-IDENTICAL REPLAY

MAUSAM[AI] · DESIGN

Mock runs offline on captured IMD payloads; the keyed gateway is the Day-1 critical path.

Analysis of the feasibility of the idea

The mock runs offline on real captured IMD payloads. **32 tests green**. Each risk below has a working fallback.

Potential challenges and risks



Keyed API needs a static-IP host



Run on saved captures + public WFS meanwhile



CPCB feed may be unreachable



Labelled mirror fallback, else an honest gap card



Some IMD fields are undocumented



Documented fallbacks; the screen states its data grain



iOS limits push alerts



Demo on Android; the inbox keeps every alert

Day 1 · static-IP host + key registration

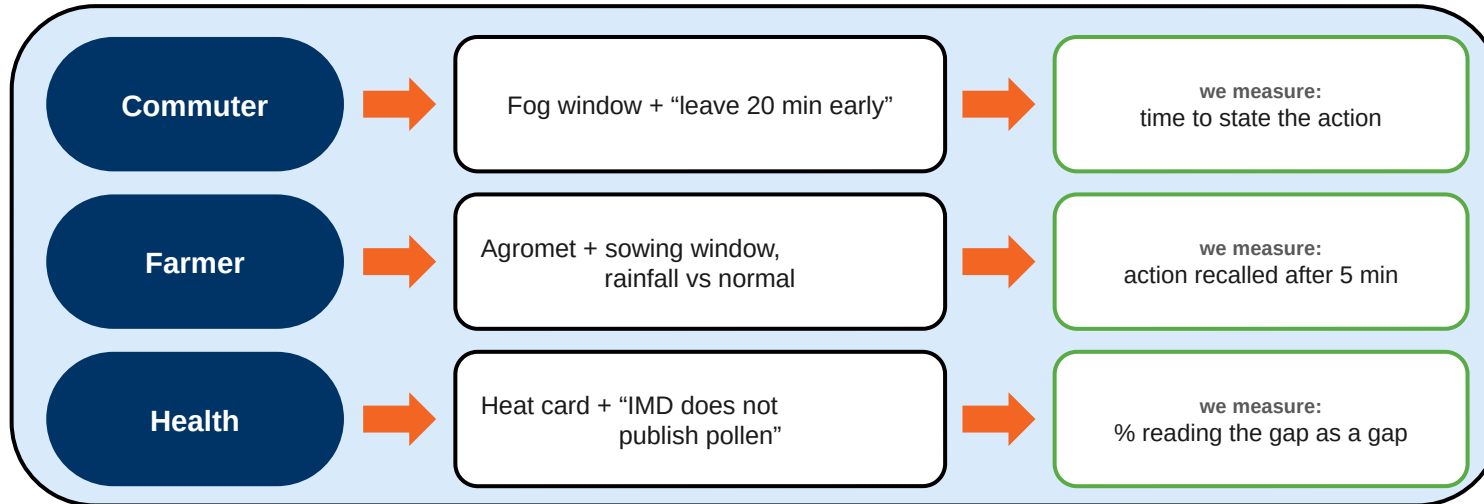
Week 1 · captured-payload mock
(done)

Week 3 · live keyed pipeline

Finale · demo: 3 personas, push,
adapters

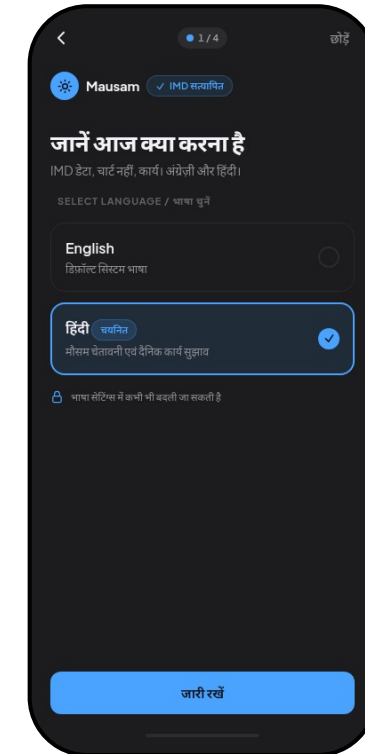
Viability: one container on a static-IP VM · one IMD API account (2 DEV + 2 PROD keys, attribution ledger on request) · no paid services · IMD data with attribution; CPCB/CAMS labelled adapters.

Potential impact on the target audience



Benefits of the solution (social, economic, environmental, etc.)

- **Earlier action:** each card states what to do and how fast.
- **Traceable trust:** every reading names source, station and age.
- **Works offline:** the home shell keeps the last forecast.
- **Hindi + English:** warning text is human-reviewed, never raw machine translation.
- **Honest gaps:** missing data shows as a gap, never a fake zero.



HINDI ONBOARDING · OFFLINE SHELL

No invented statistics. Impact is measured in the pilot.

Details / Links of the reference and research work

IMD

official · keyed API

CPCB

official · labelled adapter

Copernicus

official · labelled

MeitY UX4G

design standard

Our mock

32 tests · replay

1. IMD API reference + portal user guide (key, hourly JWT, static IP) api.imd.gov.in
2. CPCB real-time AQI, GODL data.gov.in
3. Copernicus CAMS atmosphere data store ads.atmosphere.copernicus.eu
4. MeitY UX4G design system + PIB Mausam note (2023) ux4g.gov.in · pib.gov.in
5. Pai, N. et al. (2014), IMD Pune gridded rainfall 0.25° [published dataset](#)
6. Mausam app: existing Damini lightning + Meghdoot crop modules mausam.imd.gov.in
7. Working mock, fixtures, replay tests (32 pass) [Mausam/docs/evidence \(illustrative\)](#)
8. Live demo – deployed PWA, guest-first, works offline mausam-sih26076-devanshdhangar70-7520s-projects.vercel.app/

REFERENCES

Official, verifiable sources; local mock paths are labelled (illustrative). No fabricated statistics.